// STRATEGY RESEARCH NOTE

RELATIVE STRENGTH

One intraday continuation system on NQ futures with ES as the yardstick. At 09:00 CT it checks whether NQ has broken away from what its ES beta explains since the open, and trades NQ in the direction of that break.

INSTRUMENT	SAMPLE	PERIOD	TIMEFRAME
NQ Futures	**1,290 trades**	**2017 / 2026**	**1 Minute**
DIRECTION	REFERENCE	PLATFORM	SETUP
Long & Short	**ES Futures**	**MultiCharts**	**1, one trade a day**

// CONTENTS

CONTENTS.

01	**Research Origin** ONE PROBE, TWO PAPERS	03
02	**Strategy Specification** RULES, ONE TRADE A DAY	04
03	**Headline Performance** KPIS AND FULL REPORT	05
04	**Equity and Drawdown** CURVE, UNDERWATER, MONTHLY	06
05	**Trade Distribution** PAYOFF, EXITS, DIRECTION	07
06	**In-Sample vs Out-of-Sample** 2023 BOUNDARY SPLIT	08
07	**Prop-Firm Pass Rate** TIP FUNDED SIM, APEX 50K	09
08	**Robustness** 0.5 R:R VERSION, PARAMETER SWEEPS	10
09	**Verdict** STATUS OF THE MODEL	11

// ABSTRACT

THE CASE.

Relative Strength is an intraday system on NQ futures built around one idea: **when NQ breaks away from ES in the first half hour, the break tends to carry into the afternoon**. It trades long and short, one trade a day.

At **09:00 CT** the system takes NQ's move since the 08:30 open, removes the part that ES's move explains through a rolling beta, and scales what is left by its normal size. If that relative move is large enough, it trades NQ in the same direction, with a stop and a target tied to the average daily range, and closes by 15:00 CT.

Across **1,290 trades** from January 2017 to October 2026 the MultiCharts run shows **$377,945** net profit on one NQ contract, a profit factor of **1.37**, a 52.0% hit rate and a **$32,325** maximum strategy drawdown, before costs. No calendar year lost money, and the out-of-sample period (2023 onward) holds the in-sample profit factor.

A **0.5 R:R version** (target 0.3, stop 0.6 x range) on the same entries is also positive in both periods, so the result does not hang on one exit setting.

In the TIP funded sim (Apex 50K evaluation) the system passes **31%** of single attempts, and **52.4%** within two independent attempts.

Outlier-Driven Shorts

Shorts earn $153,080 overall, but MultiCharts' outlier-adjusted figure for the short side is ($5,025). The short side lives on its largest down days; the long side does not need its outliers ($192,050).

01

// RESEARCH AND TRANSLATION

WHERE IT COMES FROM.

// Starting Point · Own Lead-Lag Probe

NER1 probe, NQ and ES 1-minute data, 2017 to 2022. At this resolution NQ leads ES. After an NQ/ES divergence, fading NQ lost 1 to 2 basis points, and ES moved 0.6 to 2.6 basis points toward NQ within 15 minutes. The divergence carries information, and it points in NQ's direction.

The probe stopped at 15 minutes. The open question was whether the relative move keeps going for hours, long enough to trade with one contract and a daily range stop.

// Supporting Paper · Intraday Momentum

Gao, L., Han, Y., Li, S. Z. and Zhou, G. (2018). "Market Intraday Momentum." *Journal of Financial Economics* 129(2), 394-414.

> *Based on high frequency S&P 500 exchange-traded fund (ETF) data from 1993-2013, we show an intraday momentum pattern.*
>
> GAO ET AL., JFE 129, ABSTRACT

The first half-hour return of the day predicts the last half-hour return. The effect is stronger on volatile days, high-volume days and macro news days, and it also shows up in ten other heavily traded ETFs. The authors link it to infrequent portfolio rebalancing and late-informed trading near the close.

// Supporting Paper · Hedging Flows

Baltussen, G., Da, Z., Lammers, S. and Martens, M. (2021). "Hedging Demand and Market Intraday Momentum." *Journal of Financial Economics* 142, 377-403.

> *Hedging short gamma exposure requires trading in the direction of price movements.*
>
> BALTUSSEN ET AL., JFE 142, P. 377 (ABSTRACT)

Across more than 60 futures from 1974 to 2020, the return of the day up to the last half hour predicts the direction into the close. Option market makers and leveraged ETFs hedge in the direction of the move, which makes it persist. NQ is part of the sample and among its strongest results.

// Translation Into The System

Both papers look at each market on its own: the early move predicts the rest of the day. This system applies the idea to the part of NQ's move that ES does not explain. If NQ-specific buying or selling dominates the first half hour, the persistence should show in **NQ's lead over ES**. The system measures that lead at 09:00 and **trades NQ in its direction**.

The trade is NQ outright, not a spread, so it also carries the market's own direction for the rest of the day.

// WHAT THIS IS NOT

The papers describe market-wide intraday momentum, not this rule. The relative-move score, the 09:00 check, the threshold and the exits are design choices tested on NQ/ES 1-minute data. The probe and the papers explain *why* the setup is built this way; they do not test it.

02

// AS IMPLEMENTED

SPECIFICATION.

// Signal

- Beta of NQ on ES and the residual variance: exponential averages over 20 days of 1-minute RTH bar returns in basis points. NYSE holidays and early-close sessions are left out.
- Relative move = NQ return since the 08:30 CT close minus beta x ES return over the same span.
- Score Z = relative move divided by its expected size: residual 1-minute variance x minutes since 08:30, square-rooted.
- At 09:00 CT: Z at or above +1.25 buys NQ, Z at or below -1.25 sells NQ short, market order on the next bar. Days in between are skipped. One trade a day at most.

// Exits

- Stop 0.6 x and target 1.0 x the average 08:30-15:00 range of the last 20 sessions, both measured from the fill.
- Flat at 15:00 CT on the bar close. No overnight positions.

// Why ES

ES is the yardstick, not a traded leg. Taking out the part of NQ's morning that is just the market moving leaves a signal that reacts to NQ-specific strength or weakness. Trading against the relative move (betting that the laggard catches up) is the exact mirror on NQ and loses.

// Inputs As Tested

Check time	09:00 CT
Z threshold	±1.25
Beta / variance lookback	20 days
Stop	0.6 × avg range
Target	1.0 × avg range
Range lookback	20 sessions
Flat	15:00 CT
Size	1 contract

FIG. A STRATEGY INPUTS (RELSTRENGTH_V3)

// Data & Test Environment

Symbol	NQ continuous (@NQ#C)
Reference series	ES continuous (@ES#C), Data2
Chart	1 Minute, both series
Session / time zone	24h Globex, CT
First / last trade	2017-01-10 / 2026-10-13
Initial capital	$100,000
Commission / slippage	$0 / $0 (not included)
Point value	$20
Intrabar order generation	Off

FIG. B ENVIRONMENT AS RECORDED IN THE WORKBOOK

03

// ANALYSING THE RESULT

DOES IT MAKE MONEY?

// NET PROFIT	// PROFIT FACTOR	// MAX STRATEGY DD	// ANN. SHARPE
$377,945	**1.37**	**$32,325**	**1.18**
1,290 trades, before costs	adj. 1.27 / select 1.30	intraday peak-to-trough, Jan 2026	monthly basis, MultiCharts

PERFORMANCE SUMMARY	ALL TRADES
Net Profit	**$377,945**
Gross Profit	$1,399,530
Gross Loss	($1,021,585)
Adjusted Net Profit	$282,789
Long Net Profit	$224,865
Short Net Profit	$153,080
Max Strategy Drawdown	($32,325)
Max Close-to-Close DD	($27,950)
Return on Max Strategy DD	**11.69**
Largest Winner	$10,440
Largest Loser	($6,825)

RATIOS AND RATES	ALL TRADES
Profit Factor	**1.37**
Percent Profitable	**52.02%**
Avg Trade	$292.98
Avg Winner / Avg Loser	$2,086 / ($1,656)
Avg Monthly Return	$3,203
Monthly Return StdDev	$8,386
Sharpe / Sortino (monthly)	0.34 / 0.69
Max Consecutive Winners / Losers	13 / 10
Positive Months	75 of 118 (63.6%)
Best / Worst Month	$29,840 / ($14,770)
Total # of Trades	**1,290**

// TIME IN MARKET

Trading period 9 years, 9 months. Position held only **7.2%** of elapsed time: one trade a day at most, 09:01 to 15:00 CT.

// COSTS

The run has **no commission or slippage**. At $5 + $1 per trade, as in our other notes, costs come to **$7,740**: net $370,205, average trade $287.

// DIRECTION

662 long and 628 short trades. Longs earn $224,865 (PF 1.46), shorts $153,080 (PF 1.29). Both sides pay.

04

// VISUAL INSPECTION

EQUITY / DRAWDOWN.

The curve climbs from $100k to roughly $478k on a fixed single contract. No calendar year lost money; 2017, 2019 and 2023 were the flattest. The steeper slope after 2023 follows NQ's larger daily ranges, which the range-based exits scale with.



FIG. 1 EQUITY CURVE, CLOSED TRADES, $100K BASE, ONE NQ CONTRACT, JAN 2017 TO OCT 2026. DASHED LINE: 2023 IN-SAMPLE / OUT-OF-SAMPLE BOUNDARY

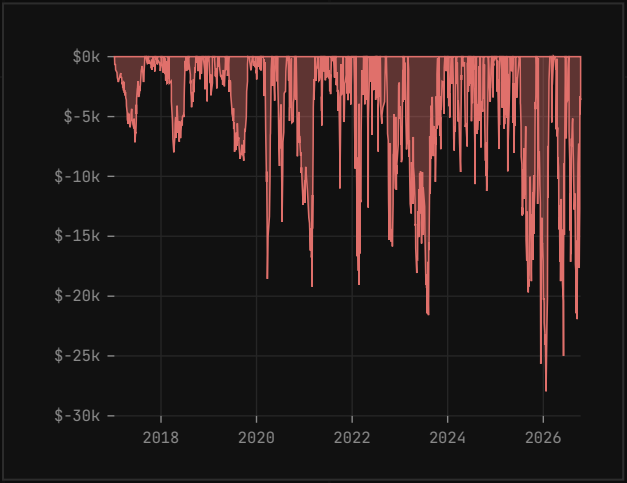


FIG. 2 UNDERWATER CURVE, CLOSED TRADES

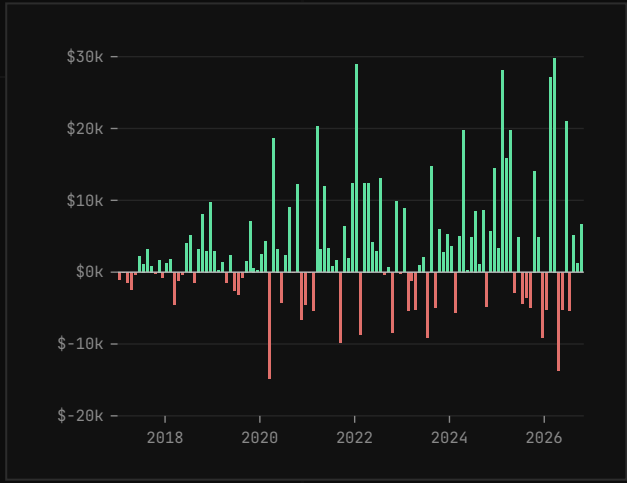


FIG. 3 NET PROFIT PER MONTH

// DRAWDOWN ENVELOPE

The deepest drawdown is recent: **January 2026**, **$32,325** intraday and $27,950 closed. It sits in the out-of-sample period; the curve made a new high again on 20 February 2026.

// MONTHLY PROFILE

75 of 118 months positive (63.6%). Best month $29,840 (Mar 2026), worst ($14,770) (Mar 2020), average month $3,203.

05

// PER-TRADE GEOMETRY

DISTRIBUTION.

Across 1,290 trades the system wins **52.0%** of the time with an average winner of **$2,086** against an average loser of **$1,656**. This is a payoff system, not a hit-rate system: the edge comes from winners that are larger than losers. The median trade is **$80**.

P&L BUCKET	TRADES
Below -$5,000	13
-$5,000 to -$2,500	142
-$2,500 to -$1,000	189
-$1,000 to $0	273
$0 to $1,000	**232**
$1,000 to $2,500	234
$2,500 to $5,000	146
Above $5,000	61

DERIVED FROM THE EXPORTED TRADE LOG

EXIT	TRADES	NET P&L	AVG
Target	151	$592,990	$3,927
Timed flat	829	$541,990	$654
Stop loss	310	($757,035)	($2,442)

DIRECTION	TRADES	NET P&L	PF
Long	662	$224,865	1.46
Short	628	$153,080	1.29

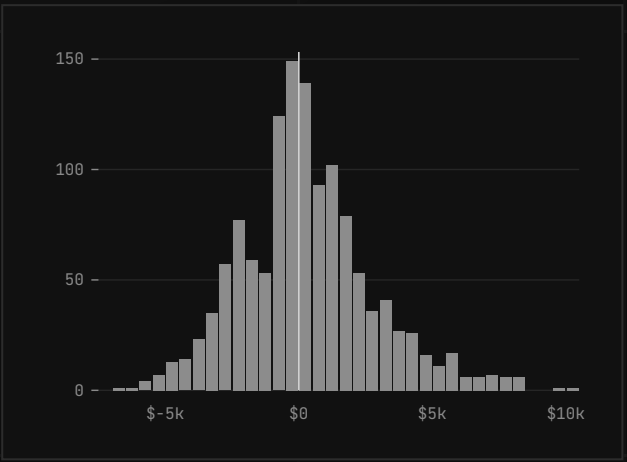


FIG. 4 PER-TRADE P&L DISTRIBUTION, $500 BINS

// CONCENTRATION

The largest winner, $10,440, is 2.8% of net profit; the five largest together are 11.9%. Removing them still leaves **$333,105**. No single trade carries the result.

The afternoon carries it. 829 of 1,290 trades (64%) are still open at 15:00 and close there at $654 on average. Targets are rare (151 trades) but large; stops end 310 trades (24%).

Both directions earn. Longs earn more and win more often (56.9% against 46.8%), so part of the result comes from NQ's rising decade, but the short side is profitable on its own.

Long holding time. The median trade lasts 359 minutes: most positions run from the 09:01 entry to the 15:00 flat.

06

// DEVELOPMENT PERIOD VS LATER PERIOD

SPLIT SAMPLE.

Development and parameter selection were carried out on **January 2017 to December 2022**. The following period, **January 2023 to October 2026**, was used afterwards to check whether the logic held outside the development window. The split is computed directly from the exported trade log.

METRIC	IN-SAMPLE	OUT-OF-SAMPLE
Period	2017-2022	2023-Oct 2026
Trades	**728**	**562**
Net profit	**$174,725**	**$203,220**
Profit factor	**1.40**	**1.35**
Win rate	**52.75%**	**51.07%**
Avg trade	$240	$362
Avg winner	$1,595	$2,743
Avg loser	($1,279)	($2,124)
Max closed-trade DD	($19,215)	($27,950)
Largest loser	($5,055)	($6,825)
Sharpe, annualized	1.23	1.49

DRAWDOWN IS CLOSED-TRADE; SHARPE FROM MONTHLY P&L

The out-of-sample period holds. Profit factor 1.40 in-sample and 1.35 out-of-sample, win rate 53% against 51%. The average trade rises from $240 to $362, but the average winner and loser grow with it: NQ trades at a higher price and moves more points per day since 2023, and the range-based exits scale with it. Profit factor and win rate are the fair comparison.

Two qualifications. This is a single chronological split, not a walk-forward. And this run is one continuous chart from 2017; a separate run that started on 1 January 2023 showed 555 trades and $208,765, the small gap coming from the warm-up of the rolling averages.

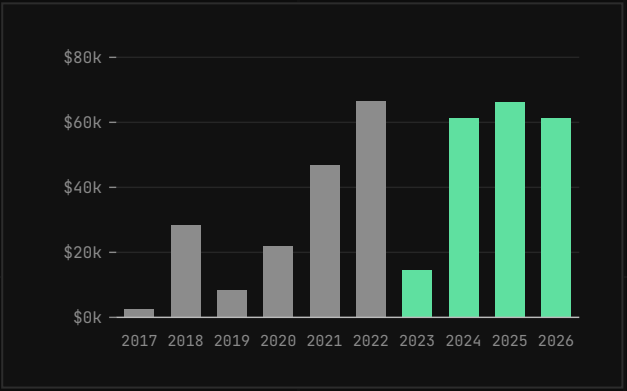


FIG. 5 NET PROFIT PER YEAR. GREY IN-SAMPLE, GREEN OUT-OF-SAMPLE

YEAR	TRADES	NET P&L	PF	WIN %
2017	117	$2,630	1.11	42.7
2018	122	$28,520	1.59	51.6
2019	113	$8,275	1.22	45.1
2020	124	$21,950	1.23	62.1
2021	124	$46,750	1.58	58.9
2022	128	$66,600	1.43	54.7
2023	148	$14,580	1.11	48.0
2024	144	$61,245	1.47	53.5
2025	155	$66,140	1.37	50.3
2026 (to Oct)	115	$61,255	1.43	53.0

NO LOSING CALENDAR YEAR; 75 OF 118 MONTHS POSITIVE (63.6%)

07

// TIP FUNDED SIM

PROP-FIRM PASS RATE.

The exported trade log was run through the **TIP funded sim**, our Monte Carlo of the **Apex 50K evaluation**. The system passes **31%** of single attempts. The attempt figures below are derived from that rate, once for independent attempts and once for attempts that share a common fate (rho 0.3).

// PASS, 1 ATTEMPT	// WITHIN 2	// WITHIN 3	// AVG ATTEMPTS
31%	**52.4%**	**67.1%**	**3.2**
TIP funded sim, Apex 50K	independent attempts	independent attempts	to the first pass (1 / 31%)

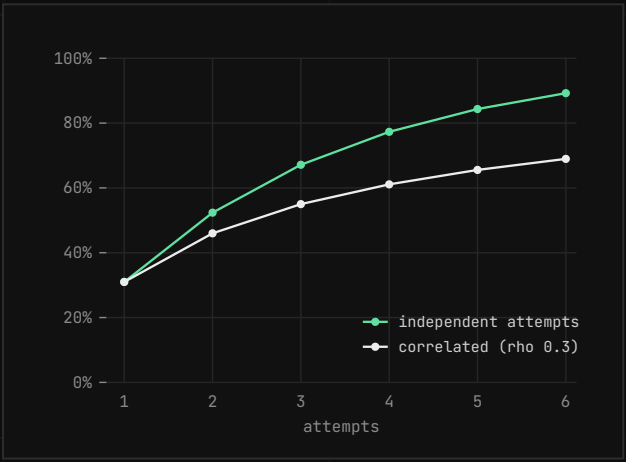


FIG. 6 PROBABILITY OF PASSING WITHIN N ATTEMPTS

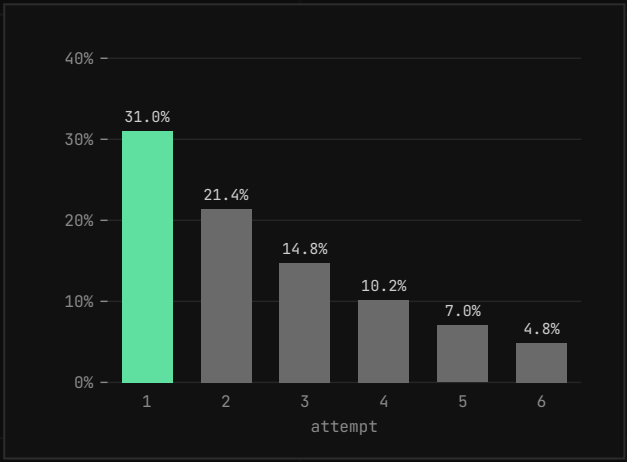


FIG. 7 PROBABILITY THAT ATTEMPT N IS THE FIRST PASS

ATTEMPTS	INDEPENDENT	CORRELATED
1	31.0%	31.0%
2	52.4%	46.0%
3	67.1%	55.0%
4	77.3%	61.1%
5	84.4%	65.5%

CORRELATED: ATTEMPTS SHARE A COMMON FATE, RHO 0.3

Reading it: about one evaluation in three passes. On average it takes 3.2 attempts to pass once; within three attempts the chance is 67.1%, or 55.0% if the attempts share a common fate.

Why not higher: this is a payoff system, not a hit-rate system. The average loser, about $1,656 per NQ, is close to the $2,000 trailing drawdown, so one or two stops end most evaluations.

Exit choice does not fix it: the 0.5 R:R version wins two trades in three, but its losers are just as large ($1,783 on average), so the same drawdown limit binds.

08

// DOES IT HANG ON ONE SETTING?

ROBUSTNESS.

// Same Entries, 0.5 R:R

The entry does not depend on one exit setting. With the target cut to **0.3 x range** and the stop left at **0.6 x range** (a 0.5 reward-to-risk ratio), the same 1,290 entries stay profitable, in-sample and out-of-sample.

METRIC	FROZEN	0.5 R:R
Stop / target, x range	0.6 / 1.0	0.6 / 0.3
Net profit	$377,945	$224,670
Profit factor	1.37	1.28
Win rate	52.0%	65.3%
Avg trade	$293	$174
Avg winner	$2,086	$1,213
Avg loser	($1,656)	($1,783)
Max strategy DD	($32,325)	($29,260)
Ann. Sharpe (MC)	1.18	0.93
In-sample net	$174,725	$99,920
In-sample PF	1.40	1.29
Out-of-sample net	$203,220	$124,750
Out-of-sample PF	1.35	1.27
Short net, no outliers	($5,025)	$57,170

BOTH FROM MULTICHARTS EXPORTS, JAN 2017 TO OCT 2026, NO COSTS

The 0.5 R:R version wins about two trades in three and banks moves early; the frozen build lets the afternoon run and earns more per trade. The quick target also makes the short side hold up without its outliers ($57,170 against ($5,025)). Its flat years are 2017 and 2019, at ($15) and ($310).

YEAR	FROZEN	0.5 R:R
2017	$2,630	($15)
2018	$28,520	$13,215
2019	$8,275	($310)
2020	$21,950	$28,960
2021	$46,750	$20,725
2022	$66,600	$37,345
2023	$14,580	$17,605
2024	$61,245	$23,170
2025	$66,140	$47,930
2026	$61,255	$36,045

NET P&L PER YEAR, ONE NQ CONTRACT

// One Dial Per Run

Every input was swept once on 2017-2022 with the others fixed, and picked from a plateau:

- **Check time x threshold:** at 09:00, Z from 1.0 to 2.0 all give $212 to $261 per trade.
- **Beta:** regression, vol ratio or fixed 1.0 over 10 to 60 days: $197 to $233, flat.
- **Entry timing:** entering on the first crossing later in the morning added 400 to 480 trades at about $0 net.
- **Market gate:** trades with ES's direction ($247) and against it ($213) both pay; no gate.
- **Exits:** stops from 0.5 to 1.0 x range form a flat plateau; targets wider than 0.75 x range match no target (sweep run with a $3,000 daily cap).

09

// STATUS OF THE MODEL

VERDICT.

Validated / Pre-Cost

One continuation system with one setup, 1,290 trades over almost ten years, no losing year, and an out-of-sample profit factor that matches the in-sample one. It earns through afternoon follow-through on both sides; the short side depends on its largest moves.

// WHAT HOLDS UP

Profit factor 1.35 out-of-sample against 1.40 in-sample.

Profitable long and short, no losing calendar year.

The 0.5 R:R version on the same entries is also positive.

// WHAT TO WATCH

Short side flat once outliers are removed.

Deepest drawdown $32,325, January 2026.

Dollar risk per trade grows with volatility.

// WHAT IS MISSING

No costs in the run.

No walk-forward.

No tick-level fill reconstruction.

No test on other index pairs.

// References

Gao, L., Han, Y., Li, S. Z., Zhou, G. (2018). Market Intraday Momentum. *Journal of Financial Economics* 129(2), 394-414.

Baltussen, G., Da, Z., Lammers, S., Martens, M. (2021). Hedging Demand and Market Intraday Momentum. *Journal of Financial Economics* 142, 377-403.

// Statement Of Limitations

Not financial advice. This research note is for educational purposes only. It shares an idea and how it was built so that others can study it, work on it and develop something of their own. It is not a finished product and does not constitute an offer or recommendation to trade any instrument. All figures derive from MultiCharts backtest exports on one NQ contract with a $100,000 nominal balance; commission and slippage are not included. Backtested results are hypothetical, carry the benefit of hindsight and do not represent actual trading. Prop-firm pass rates are simulations from resampled history. Past performance, backtested or realised, does not indicate future results.